DEPARTMENT OF COMPUTER APPLICATIONS
PSG COLLEGE OF TECHNOLOGY
MCA G1 & G2, SEMESTER 1
26MX16 - PROGRAMMING AND DATA STRUCTURES LABORATORY
PRACTICE SHEET 8

Prerequisite	C fundamentals, data types, variables, input/output operations, operators, conditional statements, loops, arrays, functions and structure basics.

1. A college maintains the academic records of its students. Each student has a register number, name, and marks in five subjects. The college wants to calculate the academic performance of each student. Write a C program using array of structure to store the register number, name and marks in five subjects. Perform the following using a user defined function.

- Calculate the total and average marks.
- Calculate the percentage.
- Display the grade based on the following criteria:
  - ✓ 90 and above – O
  - ✓ 80–89 – A+
  - ✓ 70–79 – A
  - ✓ 60–69 – B+
  - ✓ 50–59 – B
  - ✓ Below 50 – F
- Display the student details along with the results.

2. An online shopping application maintains product details such as product ID, product name, quantity purchased and unit price. Write a C program using a structure to store details of five purchased products. Perform the following using a user defined function.

- Calculate the cost of each product.
- Calculate the total purchase amount.
- Provide a 10% discount if the total purchase amount exceeds ₹5,000.
- Calculate GST at 5% on the discounted amount.
- Display the final invoice with product-wise and overall calculations.

3. A hospital maintains patient details, including patient ID, patient name, number of consultation days, and consultation fee per day, medicine charges and laboratory charges. Write a C program using a structure to store the details of five patients. Perform the following using a user defined function.

- Calculate consultation charges.
- Calculate the total medical expenses.
- Provide a 5% discount if the total bill exceeds ₹10,000.
- Calculate the final payable amount.
- Display the patient-wise bill details.

4. A vehicle rental company maintains vehicle details such as vehicle number, vehicle type, customer name and number of rental days. The rental charges are calculated as follows:

Vehicle type	**Daily rental charge**
Two-wheeler	₹ 500
Hatchback	₹ 1,200
Sedan	₹ 1,800
SUV	₹ 2,500

Write a C program using a structure to store the vehicle and customer details. Perform the following using a user defined function.

- Calculate the rental amount based on vehicle type and rental days.
- Add a security service charge of ₹200.
- Provide a 10% discount for rentals exceeding seven days.
- Display the rental bill.

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5. A retail company maintains salesperson details, including employee ID, salesperson name and total sales achieved during a month. The company provides commission based on sales:

Monthly sales	Commission
Below ₹25,000	2%
₹25,000–₹50,000	5%
₹50,001–₹1,00,000	8%
Above ₹1,00,000	10%

Write a C program using a structure to store the employee ID, name, and monthly sales. Perform the following using a user defined function.

- Calculate the commission based on the sales achieved.
- Calculate the total earnings assuming a fixed salary of ₹15,000.
- Display the salesperson details, sales amount, commission, and total earnings.

6. A transport company maintains passenger booking details such as ticket number, passenger name, source, destination, distance travelled and passenger category. The fare is calculated at ₹2 per kilometre. Discounts are provided based on passenger category:

- Child – 50% discount
- Senior citizen – 30% discount
- Regular passenger – No discount

Write a C program using a structure to store the passenger booking details. Perform the following using a user defined function.

- Calculate the basic ticket fare based on distance.
- Calculate the discount based on passenger category.
- Calculate the final ticket fare.
- Display the passenger details and fare statement.

Skills Acquired	Structure declaration and initialization, structure variables, member access, array of structures, structure-based calculations, conditional processing, record management, data organization, and application-oriented problem solving.